

1. Without using the simplex algorithm, justify the optimal solution of the LP shown below.

$$\begin{aligned} \text{Max } Z &= 4x_1 + 6x_2 + 7x_4 \\ \text{subject to} \\ 3x_1 + 2x_2 + x_3 + 4x_4 &= 1 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

2. Solve the following LP using x_3 and x_4 as the starting basic variables and without using any artificial variables.

$$\begin{aligned} \text{Minimize } z &= 3x_1 + 2x_2 + 3x_3 \\ \text{subject to} \\ x_1 + 4x_2 + x_3 &\geq 7 \\ 2x_1 + x_2 + x_4 &\geq 10 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$